Experiment No. 6

Parseval's Relation and Harmonic Spectrum Analysis

OBJECTIVE:

- To verify Parseval's relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-domain power.
- To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.

I. To verify Parseval's relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-domain power.

Experimental Procedure:

- Generate the periodic signal over one period.
- Compute time-domain average power.
- Calculate Fourier coefficients up to N = 20.
- Evaluate frequency-domain power using Parseval's relation.
- Compare the two values and calculate absolute error.

MATLAB Coding:

exp_6a.m

/MATLAB Drive/exp_6a.m

```
clc;
clear;
close all;

figure;

f0 = 1;
T = 1/f0;
t = linspace(0,T,4000);

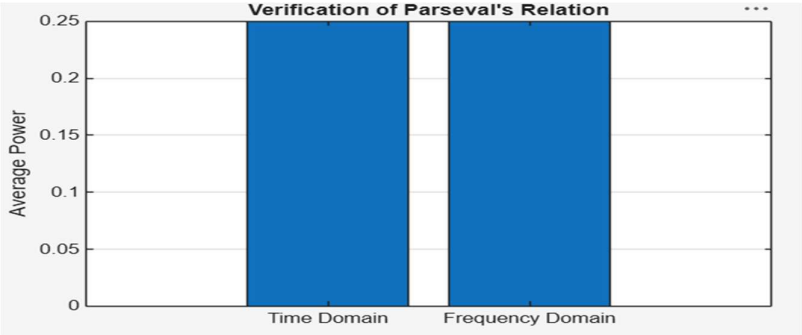
x = max(sin(2*pi*f0*t),0);

% Time-domain average power
P_time = (1/T)*trapz(t,abs(x).^2);

% Fourier coefficients
N = 20;
a0 = (2/T)*trapz(t,x);
an = zeros(1,N);
bn = zeros(1,N);

for k = 1:N
    an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
end
```

Results:



II. To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.

Experimental Procedure:

- Generate the periodic signal.
- Compute a0, an and bn up to N = 20.
- Calculate power contribution of each harmonic.
- Plot harmonic number versus power contribution.
- Record the DC contribution and dominant harmonics.

MATLAB Coding:

```
clc;
clear;
close all;

figure;

f0 = 1;
T = 1/f0;
t = linspace(0,T,4000);

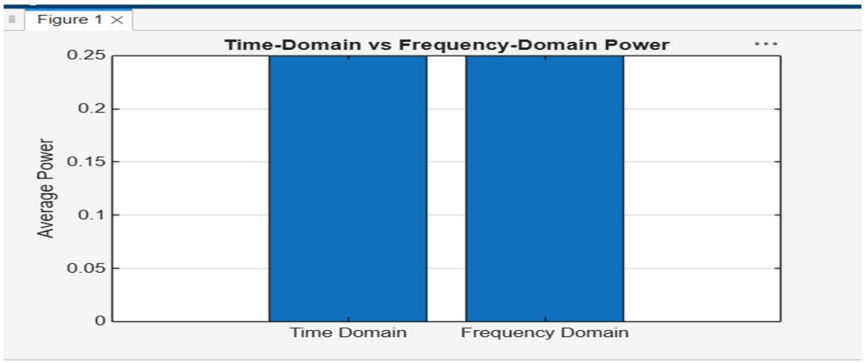
x = max(sin(2*pi*f0*t),0);

% Time-domain power
P_time = (1/T)*trapz(t,x.^2);

% Fourier coefficients
N = 20;
a0 = (2/T)*trapz(t,x);
an = zeros(1,N);
bn = zeros(1,N);

for k = 1:N
    an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
```

Results:



Conclusion:

The experiment successfully verified **Parseval's relation** for continuous-time periodic signals by demonstrating the equivalence of power calculated in the time and frequency domains. The distribution of power among the fundamental and harmonic components was analyzed through the **harmonic spectrum**, and the total time-domain power was found to be consistent with the sum of harmonic powers in the frequency domain.